Compare Linear Classification Models on Spam Detection and English & Dutch Language Identification

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1 Introduction

This project aims to investigate the performance of different linear classification models on spam detection and English-Dutch language identification. I implemented and compared three models: Perceptron, Logistic Regression, and Linear Support Vector Machine (Linear SVM).

For the spam detection task, I used the Spambase dataset, which contains numerical features and corresponding labels. For the English-Dutch language identification task, I used texts from the Universal Declaration of Human Rights and other English and Dutch texts, which contain text data. By comparing the models, tuning hyperparameters, and experimenting with different text representations, I investigated which methods work best for each task.

2 Methods

2.1 Linear Classification Models

2.1.1 Perceptron

I built perceptron using NumPy and scikit-learn. Perceptron will update the weight and bias when it has classification errors. I experimented a random state and several stopping criterion to investigate the better performance.

2.1.2 Logistic Regression

A linear classifier that models class probabilities using the logistic function. I set maximum number of passes over the training data `max_iter=5000` to allow sufficient iterations for optimization.

2.1.3 Linear Support Vector Classifier (LinearSVC)

A linear support vector classifier that learns a decision boundary by maximizing the margin between classes. I used `tol=1e-5` in the experiments.

2.2 Spam Detection

I used the training set to train the models and the development set to evaluate their performance and tune hyperparameters based on accuracy. I used the test set to assess the models' generalization ability. All three models were trained and evaluated using the same training and development sets, and the same test set was used to compare their performance. I trained all three models on the same training set. For Perceptron, I used `tol=None` after an initial experiment since `tol=1e-3` produced obviously lower accuracy. Logistic Regression used `max_iter=5000`, and LinearSVC used `tol=1e-5` with feature standardization through a pipeline.

2.3 English & Dutch Language Identification

I combined the English and Dutch training texts and assigned language labels. The development and test texts came from CNN and DutchNews articles and *The Discovery of Heaven*. I converted the text into numerical features using `CountVectorizer` with `analyzer='char'` and `ngram_range=(2, 3)`, which counts character sequences of two and three characters. I fitted the vectorizer on the training data and used it to transform the development and test data. Finally, I trained Perceptron, Logistic Regression, and LinearSVC using these features.

3 Result & Discussion

3.1 Spam Detection

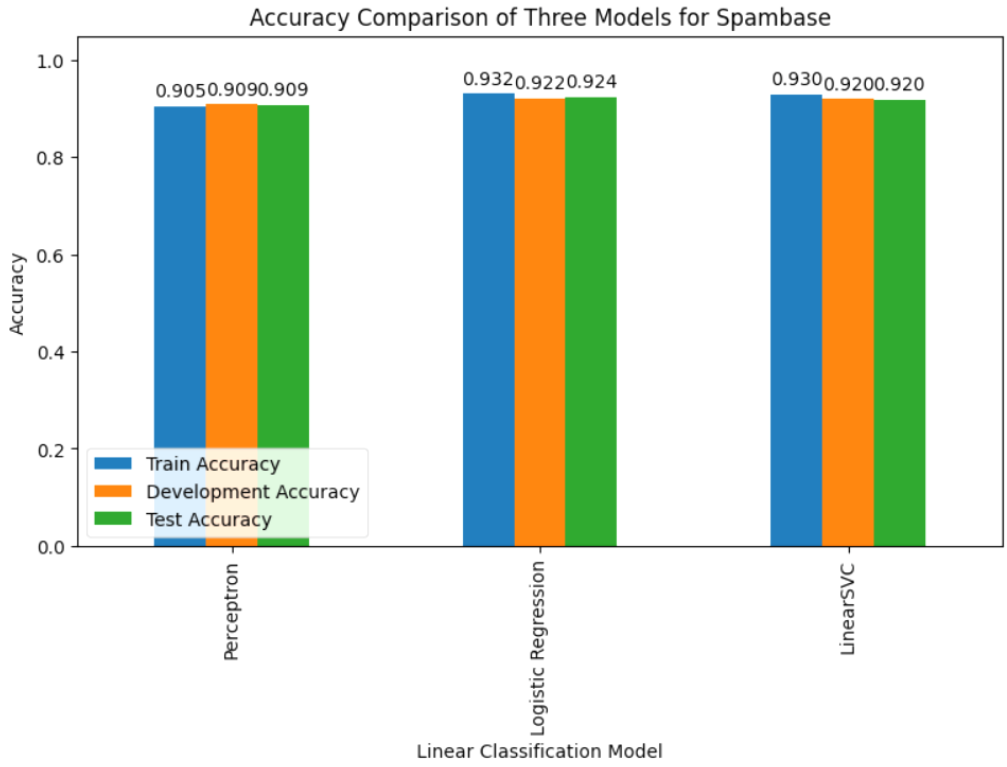


Figure 1: Bar chart of Accuracy Comparison of Three Models for Spambase

As shown in the bar chart generated from the visualization, Logistic Regression has the highest test accuracy at 0.924, followed by LinearSVC and Perceptron. This suggests that Logistic Regression performed best among the three models for spam detection in these experiments.

The training and development accuracies were relatively close for all three models. In particular, Perceptron's development accuracy was slightly higher than its training accuracy. But the differences between the training, development is slight, test results do not indicate a large generalization gap in these experiments.

Perceptron's performance also improved considerably after I changed its tolerance setting from `1e-3` to `None`. Its initial development accuracy was approximately 57.92%, whereas the later experiment achieved approximately 90.89%. This shows that model configuration can affect the results obviously.

3.2 English & Dutch Language Identification

All three models achieved 100% training and development accuracy, while Logistic Regression and LinearSVC also achieved 100% test accuracy.

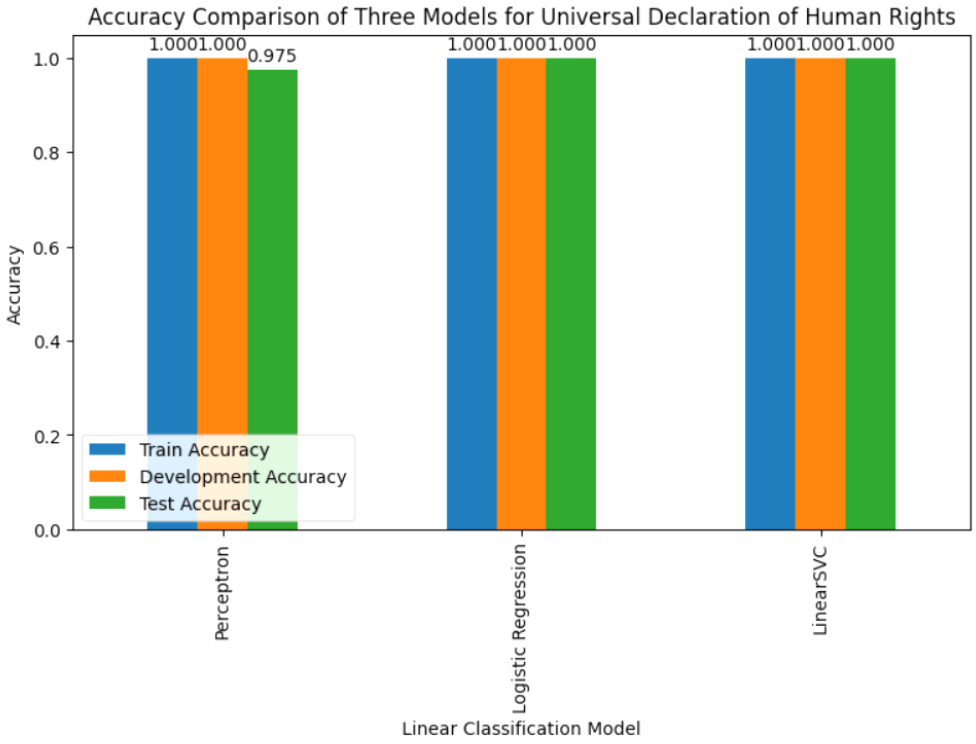


Figure 2: Caption

The results suggest that character n-gram features were effective for distinguishing English from Dutch. Individual characters and short character sequences provide information about spelling patterns and character frequencies. However, the perfect training accuracy and lower test accuracy suggest that the models may be fitting some patterns that do not generalize equally well to unseen text.

Perceptron had the largest gap between training and test accuracy. Logistic Regression and LinearSVC generalized better on the test set in this experiment. Therefore, LinearSVC was the strongest of the three models for the current language-identification setup.

3.3 Features tried

I experimented with TF-IDF features and CountVectorizer with different n-gram ranges to distinguish English from Dutch.

I first used TF-IDF as a baseline. All three models achieved 100% training and development accuracy and 97.5% test accuracy.

I experimented with different combinations of unigrams, bigrams, and trigrams. With the (1, 2) configuration, the test accuracies were 90% for Perceptron, 87.5% for Logistic Regression, and 90% for LinearSVC. With the (1, 3) configuration, the test accuracies were 85% for Perceptron, 92.5% for Logistic Regression, and 95% for LinearSVC in the corresponding experiments. I then excluded unigrams and used only bigrams and trigrams. The test accuracies reached 97.5% for Perceptron, 100% for Logistic Regression, and 100% for LinearSVC.

These results suggest that bigrams and trigrams can provide useful character patterns for distinguishing English from Dutch. Excluding unigrams also improved the observed test accuracy in these experiments. But this does not establish that (2, 3) is always the best configuration.

3.4 Stress Testing

All three models achieved 100% training and development accuracy on the English-Dutch language identification task, possibly because of the distinct differences between the two languages. However, we need to stress-test the models.

3.4.1 Shorter Inputs

Table 1: Test Accuracy on Shorter Text Fragments

Fragment Length	Perceptron	Logistic Regression	LinearSVC
1 word	64.62%	64.51%	69.88%
2 words	68.38%	76.52%	78.01%
3 words	73.35%	85.35%	84.15%
5 words	76.10%	92.43%	92.56%

As the input snippets became longer, the test accuracy of all three models increased significantly. Logistic Regression and LinearSVC consistently achieved higher test accuracy than Perceptron at every stage. This suggests that longer snippets are more helpful for distinguishing English from Dutch, and that Logistic Regression and LinearSVC are better able to make predictions based on context.

3.4.2 Learning Curve

Table 2: Development and Test Accuracy for Different Training Sizes

	Development Accuracy			Test Accuracy		
Training Size	Perceptron	Logistic Regression	LinearSVC	Perceptron	Logistic Regression	LinearSVC
2	52.5%	55.0%	100.0%	55.0%	52.5%	97.5%
4	52.5%	80.0%	95.0%	55.0%	75.0%	95.0%
10	97.5%	92.5%	97.5%	85.0%	92.5%	100.0%
20	97.5%	100.0%	97.5%	95.0%	100.0%	100.0%
40	90.0%	100.0%	97.5%	95.0%	100.0%	100.0%

All three models achieved 100% training accuracy at every training size. As the training size increased, their test accuracy improved significantly, with LinearSVC showing the largest increase and Perceptron showing the smallest. This suggests that Logistic Regression and LinearSVC can effectively distinguish English from Dutch even with relatively small training sets.

3.4.3 Check Learned Weights

The largest positive and negative coefficients from Logistic Regression and LinearSVC included character n-grams such as `ij`, `cht`, `th`, and `the`. These features are consistent with differences in English and Dutch spelling patterns, suggesting that the models learned linguistically meaningful signals. Nevertheless, individual n-grams are not exclusive to either language, and weight inspection alone cannot rule out dataset-specific artifacts.

4 Conclusion

Overall, all three models performed well on the English&Dutch classification task. Among the tested feature configurations, using bigrams and trigrams without unigrams achieved the best observed test accuracy. Logistic Regression and LinearSVC both reached 100% test accuracy, while Perceptron reached 97.5%. However, further testing would be needed to determine whether these results generalize to other datasets.